Location Intelligence & Logistics — Foundations

The domain, not the project: how space, demand, optimisation and delivery economics fit together — plus things to read and follow casually.

General-fluency primer · complements (does not repeat) the project reading list · Oct 2026 · every link verified against primary sources

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> **What this is.** The project reading list covers the specific pipeline: H3, population rasters, OSRM, maximal-covering papers, CP-SAT, the quick-commerce papers and the serving stack. This one is for the field behind it: how spatial data behaves, how retail demand is modelled, how operations researchers think, how supply chains are costed, how site selection is done in industry, and how decision tools get built and explained. Nothing here duplicates that list. §1–3 are the ideas worth genuinely internalising; §4–6 are the wider landscape and craft; §7–8 are for relaxed, ongoing absorption.

If you only do three things: (1) read **Openshaw on MAUP** and **Roberts et al.** (§1) so you distrust per-zone numbers and random train/test splits; (2) read **Wilson on spatial interaction** (§2) and **Eppen on risk pooling** (§4) to get the two core mechanisms behind store networks; (3) read **Geoffrion on insight, not numbers** (§3) and the **Saltelli manifesto** (§6), then follow one or two sources from §8 (start with **weeklyOSM** or **MIT Supply Chain Frontiers**).

1 — Spatial Thinking (why places are not rows in a table)

A Computer Movie Simulating Urban Growth in the Detroit Region — Waldo Tobler (1970), Economic Geography 46 paywalled
The home of the "first law of geography": near things are more related than distant things. Everything else in spatial statistics is a consequence or a qualification.
doi.org/10.2307/143141

Tobler's First Law and Spatial Analysis — Harvey Miller (2004), Annals of the AAG 94(2) paywalled
A careful look at what the first law does and does not license, and why it underwrites interpolation, smoothing and spatial regression.
doi.org/10.1111/j.1467-8306.2004.09402005.x

The Modifiable Areal Unit Problem (CATMOG 38) — Stan Openshaw (1984), Geo Books, Concepts and Techniques in Modern Geography
The short monograph that named the problem: change the zoning or the scale and the same data tells a different story. Read it before trusting any per-zone number.
web.archive.org/web/2011/http://qmrg.org.uk/files/2008/11/38-maup-openshaw.pdf

Notes on continuous stochastic phenomena — P. A. P. Moran (1950), Biometrika 37 paywalled
The original Moran's I paper: a single number for "are neighbours alike?". Short enough to read as a primary source.
doi.org/10.1093/biomet/37.1-2.17

Local Indicators of Spatial Association: LISA — Luc Anselin (1995), Geographical Analysis 27(2)
Decomposes global autocorrelation into per-location hot spots, clusters and outliers. The basis of most "where is the cluster" maps.
doi.org/10.1111/j.1538-4632.1995.tb00338.x

Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure — Roberts et al. (2017), Ecography 40(8) paywalled
Why random train/test splits flatter models on spatial data, and what blocked validation looks like. The cleanest statement of how spatial structure breaks i.i.d. assumptions.
doi.org/10.1111/ecog.02881

The Uncertain Geographic Context Problem — Mei-Po Kwan (2012), Annals of the AAG 102(5) paywalled
MAUP's cousin: the area you assign to a person is a guess about the environment that actually affects them. Useful whenever a hexagon stands in for a catchment.
doi.org/10.1080/00045608.2012.687349

2 — Retail & Urban Geography (how demand meets supply in space)

The Functional Bases of the Central Place Hierarchy — Berry & Garrison (1958), Economic Geography 34(2) paywalled
Christaller's central place theory made empirical: thresholds, ranges of goods and why some stores cluster while others spread out.
doi.org/10.2307/142299

Parameter Estimation for a Multiplicative Competitive Interaction Model — Nakanishi & Cooper (1974), Journal of Marketing Research 11(3) paywalled
The MCI model: how to fit Huff-style share models from data with least squares. The statistical descendant most practitioners actually use.
doi.org/10.2307/3151146

A Family of Spatial Interaction Models, and Associated Developments — Alan Wilson (1971), Environment and Planning A 3(1) paywalled
Derives gravity models from entropy maximisation and shows the origin-constrained, destination-constrained and doubly-constrained variants. The theoretical spine of spatial interaction.
doi.org/10.1068/a030001

Measures of Spatial Accessibility to Health Care in a GIS Environment (two-step floating catchment area) — Luo & Wang (2003), Environment and Planning B 30(6)
The supply-to-demand-ratio-within-a-catchment method, borrowed everywhere from health to retail. A clean way to think about capacity versus population.
doi.org/10.1068/b29120

A universal model for mobility and migration patterns — Simini, González, Maritan & Barabási (2012), Nature 484
The radiation model: an alternative to gravity that needs no fitted parameters. Good for seeing what gravity models quietly assume.
doi.org/10.1038/nature10856

Introducing the "15-Minute City": Sustainability, Resilience and Place Identity in Future Post-Pandemic Cities — Moreno, Allam, Chabaud, Gall & Pratlong (2021), Smart Cities 4(1)
The urbanist case for proximity as an organising principle. Instant delivery is the commercial mirror image of the same idea.
doi.org/10.3390/smartcities4010006

3 — Operations Research Thinking (modelling versus solving)

Model Building in Mathematical Programming (5th ed.) — H. Paul Williams (2013), Wiley paywalled
The standard book on formulation: how to turn a verbal problem into constraints and variables, including the tricks (big-M, logical conditions) that make integer models work.
wiley.com (ISBN 978-1-118-44333-0)

Optimization Methods in Management Science (15.053) — MIT Sloan, OpenCourseWare (Spring 2013)
A free course built around modelling for managers: LP, sensitivity analysis and shadow prices (the intuition behind duality), integer programs and nonlinear models.
ocw.mit.edu/courses/15-053-optimization-methods-in-management-science-spring-2013/

The Purpose of Mathematical Programming is Insight, Not Numbers — Arthur Geoffrion (1976), Interfaces 7(1) paywalled
The OR creed in six pages. A model is a way to understand trade-offs, not an oracle; the right output is a decision a manager can defend.
doi.org/10.1287/inte.7.1.81

A Brief History of Linear and Mixed-Integer Programming Computation — Robert E. Bixby (2012), Documenta Mathematica, Optimization Stories
From simplex to branch-and-cut: why integer programs were once hopeless and what changed. Gives a feel for why some formulations solve in seconds and others never do.
`doi.org/10.4171/DMS/6/16`

Heuristic "Optimization": Why, When, and How to Use It — Zanakis & Evans (1981), Interfaces 11(5) paywalled
A level-headed account of when a heuristic is the right tool and how to judge it against exact methods.
`doi.org/10.1287/inte.11.5.84`

Metaheuristics: the metaphor exposed — Kenneth Sörensen (2015), International Transactions in Operational Research 22(1)
A warning against metaphor-driven algorithms and for comparing heuristics on substance. Useful armour against hype.
`doi.org/10.1111/itor.12001`

4 — Supply Chain & Last-Mile Fundamentals (the cost structure behind the map)

Supply Chain Management: Strategy, Planning, and Operation (7th ed.) — Chopra & Meindl (2019), Pearson paywalled
The default textbook: network design, inventory, transportation, responsiveness versus efficiency. Chapters on network design and inventory pooling are the ones to read.
`pearson.com (Global Edition, ISBN 978-1-292-25789-1)`

Supply Chain Management MicroMasters — MIT Center for Transportation & Logistics, MITx on edX
The MIT SCM programme: free to audit the courses on supply chain analytics, design, fundamentals and dynamics.
`micromasters.mit.edu/scm/`

Information Distortion in a Supply Chain: The Bullwhip Effect — Lee, Padmanabhan & Whang (1997), Management Science 43(4) paywalled
Why small swings in retail demand become large swings upstream. Essential for understanding replenishment of fast-moving local inventory.
`doi.org/10.1287/mnsc.43.4.546`

Effects of Centralization on Expected Costs in a Multi-Location Newsboy Problem — Gary Eppen (1979), Management Science 25(5) paywalled
The risk-pooling result: one central stock needs less safety inventory than many local ones, by roughly the square root of the number of locations. This is the tension every dark-store network manages.
`doi.org/10.1287/mnsc.25.5.498`

Facility location and supply chain management: A review — Melo, Nickel & Saldanha-da-Gama (2009), European Journal of Operational Research 196(2) paywalled
How location decisions couple with inventory, transport and capacity. The map of the literature between textbook network design and research models.
`doi.org/10.1016/j.ejor.2008.05.007`

Last-mile delivery concepts: a survey from an operational research perspective — Boysen, Fedtke & Schwerdfeger (2021), OR Spectrum 43(1)
A structured tour of last-mile options (vans, lockers, bikes, crowdsourcing, drones) and what each does to cost and service. Open access.
`doi.org/10.1007/s00291-020-00607-8`

5 — Location Intelligence as an Industry (how site selection is done in practice)

Methods for Determining Store Trade Areas, Market Penetration, and Potential Sales — William Applebaum (1966), Journal of Marketing Research 3(2) paywalled
The founding paper of the analog method: use data from existing stores to forecast new ones. Still the template for practical site-selection work.
`doi.org/10.2307/3150201`

Retail Geography and Intelligent Network Planning — Birkin, Clarke & Clarke (2002), Wiley paywalled
A practitioner-academic account of store-network planning: spatial interaction models, data, and how retailers use them in practice.
wiley.com (ISBN 978-0-471-49803-2)

Original Huff Model (Business Analyst toolbox) — Esri, ArcGIS Desktop 10.6 documentation (archived)
How the dominant commercial GIS packages Huff's model: attractiveness, distance decay, output probabilities used to draw trade areas. See also the Huff calibration pages linked from it.
desktop.arcgis.com/en/arcmap/10.6/tools/business-analyst-toolbox/original-huff-model.htm

ArcGIS Business Analyst documentation — Esri
The vocabulary of commercial site selection: drive-time trade areas, demographic enrichment, retail gap and competitor analysis. Skim for the concepts, not the buttons.
doc.arcgis.com/en/business-analyst/web/

Linux Foundation announces Overture Maps Foundation to build interoperable open map data — Linux Foundation (Dec 2022), press release
Why Overture exists: a neutral, shared base map built from many sources, complementing OpenStreetMap rather than replacing it. Founders listed in the release.
linuxfoundation.org/press/linux-foundation-announces-overture-maps-foundation-to-build-interoperable-open-map-data

Overture Maps Foundation — Overture Maps Foundation, documentation and data releases
The current monthly releases, themes (places, buildings, transportation, addresses) and licensing. Read the places theme and its confidence field to see what open POI data does and does not offer.
docs.overturemaps.org

The world's user-generated road map is more than 80% complete — Barrington-Leigh & Millard-Ball (2017), PLOS ONE 12(8)
A quantitative check on how complete OpenStreetMap roads are, country by country. Required reading before trusting open data as ground truth.
doi.org/10.1371/journal.pone.0180698

6 — Product Engineering for Decision Tools (decisions under uncertainty, honestly communicated)

Shape Up: Stop Running in Circles and Ship Work that Matters — Ryan Singer (2019), Basecamp, free online
A small-team method: fixed time, variable scope, shaping the work before building. Pairs well with models whose scope tends to balloon.
basecamp.com/shapeup

Working Backwards: PR/FAQ — Working Backwards (Bryar & Carr), resource page
How the press-release-and-FAQ document forces clarity on who the customer is and what changes for them, before any code. Good for decision-tool briefs.
workingbackwards.com/resources/working-backwards-pr-faq/

The Visual Display of Quantitative Information (2nd ed.) — Edward Tufte (2001), Graphics Press paywalled
Data-ink, small multiples and graphical integrity: the canonical argument for showing numbers honestly. Publisher site blocks automated fetchers.
edwardtufte.com/tufte/books_vdqi

Show Me the Numbers (2nd ed.) — Stephen Few (2012), Analytics Press paywalled
The practical counterpart to Tufte: which chart for which comparison, and what to leave out of a dashboard.
perceptualedge.com/library.php

Models and Managers: The Concept of a Decision Calculus — John D. C. Little (1970), Management Science 16(8) paywalled
Argues that a model a manager can understand and adjust beats a sophisticated one they cannot. Six criteria for good decision-support models that have aged well.
doi.org/10.1287/mnsc.16.8.B466

Science and Statistics — George E. P. Box (1976), Journal of the American Statistical Association 71(356) paywalled
Source of "all models are wrong" and the less-quoted point that the question is how wrong they need to be to still be useful.
doi.org/10.1080/01621459.1976.10480949

Five ways to ensure that models serve society: a manifesto — Saltelli et al. (2020), Nature 582
Short and pointed: mind the assumptions, hubris and false precision. A checklist for what a trustworthy modelling tool must show its users.
doi.org/10.1038/d41586-020-01812-9

Communicating uncertainty about facts, numbers and science — van der Bles, van der Linden, Freeman, Mitchell et al. (2019), Royal Society Open Science 6(5)
An evidence-based framework for what to say about uncertainty, in what form, to whom. Free.
doi.org/10.1098/rsos.181870

When (ish) is My Bus? User-centered Visualizations of Uncertainty in Everyday, Mobile Predictive Systems — Kay, Kola, Hullman & Munson (2016), CHI Proceedings paywalled
Experiments on how to show uncertain predictions to ordinary people. The quantile dotplot idea transfers directly to showing a range of outcomes from a plan.
doi.org/10.1145/2858036.2858558

Decision Making under Deep Uncertainty: From Theory to Practice — Marchau, Walker, Bloemen & Popper, eds. (2019), Springer, open access
Robust decision-making and scenario discovery: how to choose when probabilities cannot be trusted. The serious version of scenario planning.
doi.org/10.1007/978-3-030-05252-2

7 — Books & Longer Reads (mostly free / open)

Geocomputation with Python — Dorman, Graser, Nowosad & Lovelace (CRC Press), free online
A hands-on modern spatial stack in Python: vector and raster handling, geometries, CRS, networks and mapping.
py.geocompx.org

Geocomputation with R (2nd ed.) — Lovelace, Nowosad & Muenchow (2025), CRC Press, free online
The original and widest of the series. The transport and location-analysis chapters are the relevant ones.
r.geocompx.org

Geographic Data Science with Python — Rey, Arribas-Bel & Wolf (CRC Press), free online
Spatial weights, autocorrelation, clustering and spatial regression in PySAL, with the intuition behind each. The practical sequel to §1.
geographicdata.science/book/

Spatial Data Science: With Applications in R — Pebesma & Bivand (2023), CRC Press, free online
A rigorous treatment of spatial data types, support, sampling and models, including the cautions on what an areal value represents.
r-spatial.org/book/

Location Science (2nd ed.) — Laporte, Nickel & Saldanha-da-Gama, eds. (2019), Springer paywalled
The reference handbook of facility location: models, solution methods and applications, chapter by chapter.
link.springer.com/book/10.1007/978-3-030-32177-2

The Power of Resilience: How the Best Companies Manage the Unexpected — Yossi Sheffi (2015), MIT Press paywalled
How firms prepare for disruption: redundancy, flexibility and culture. Gives the logistics view that cost-minimising plans can be brittle.
sheffi.mit.edu/node/91

The Magic Conveyor Belt: Supply Chains, A.I., and the Future of Work — Yossi Sheffi (2023), MIT CTL Media
A plain-language tour of how modern supply chains work and where automation is heading, from a long-time MIT CTL scholar.
sheffi.mit.edu/magicbelt

8 — Idle Learning (follow casually: active as of Oct 2026)

The MapScaping Podcast — MapScaping, podcast
Long-form interviews on geospatial technology, data and practice. Latest episode seen Oct 1, 2026.
mapscaping.com/podcasts/

weeklyOSM — weeklyOSM community, weekly digest
A weekly digest of OpenStreetMap news, tools and data-quality discussion. Latest issue seen Oct 25, 2026.
weeklyosm.eu

Free and Open Source GIS Ramblings — Anita Graser, blog
Practical writing on QGIS, movement data and open geospatial analysis by a leading practitioner. Latest post seen Oct 14, 2026.
anitagraser.com

Geoawesome — Geoawesome, news and blog
Industry-wide geospatial news: mapping, AI, earth observation and location data. Latest article seen Oct 24, 2026.
geoawesome.com

OR in an OB World — Paul Rubin, blog
A working optimiser's notes on modelling puzzles, solver quirks and formulation tricks. Latest post seen Oct 25, 2026.
orinanobworld.blogspot.com

Resoundingly Human — INFORMS, podcast
The OR and analytics society's podcast: practitioners, Edelman Award finalists and applications. Latest episode seen Oct 26, 2026 (Apple listing; the INFORMS page blocks fetchers).
podcasts.apple.com/us/podcast/resoundingly-human/id1428964002

MIT Supply Chain Frontiers — MIT Center for Transportation & Logistics, podcast
Research-led conversations on supply chains and logistics from MIT CTL. Latest episode seen Oct 20, 2026.
podcasts.apple.com/us/podcast/mit-supply-chain-frontiers/id1509776693

Talking Logistics with Adrian Gonzalez — Talking Logistics, blog and video series
Industry commentary on transportation and supply chain technology. Latest post seen Oct 26, 2026.
talkinglogistics.com

The Logistics of Logistics — Joe Lynch, podcast
A long-running, wide-ranging logistics interview show. Latest episode seen Oct 27, 2026.
podcasts.apple.com/us/podcast/the-logistics-of-logistics/id1409735368

Every link verified to resolve against a primary source (Oct 2026). DOIs were checked against Crossref for authors, year and venue. Items marked paywalled (Wiley, Sage, JSTOR, Elsevier, INFORMS, Taylor & Francis, ACM, Pearson, Springer books) block automated fetchers or sit behind subscriptions; they were confirmed real through DOI resolution and Crossref metadata, or through publisher and library catalogue records for books (ISBNs given). The Tufte and MIT Press sites return 403 to bots, so those were confirmed by search. The Openshaw monograph's original host (qmrg.org.uk) was returning a server error at check time, so the link points to a Wayback Machine copy that resolves. Idle-learning sources were checked for 2026 activity from their feeds or listings; the dates shown are the latest seen. Dropped as unconfirmed: Mike Trick's OR blog (last post 2016), the Spatial Edge newsletter (site unreachable), Logistics Viewpoints (bot-blocked, activity not confirmed), Huff and McCallum's Esri calibration white paper (its URL now redirects elsewhere; Esri's archived Huff documentation is used instead), and vendor blog posts on quick-commerce unit economics and the widely repeated "53% of shipping cost is last mile" figure (no primary source found). Peer-reviewed economics of instant delivery is still thin; Buldeo Rai et al. is the best open entry point found. Companion document: the *Catchment* project reading list.